

Chain of thoughts

Automation Framework Architecture

Context

As a Senior Test engineer create a playwright typescript framework.

Instructions

[STRICT] Create a Playwright typescript framework for UI and API testing

[MANDATORY] Generate test cases covering authentication and end cases

[STRICTLY] Ask me 5 valid questions in total. Once after answering the first question go with the next question

[MANDATORY] Proceed to create the framework based on the options received.

What is your preferred project structure approach?



1 Page Object Model (POM)



2 Screenplay Pattern

3 Simple functional approach

4 Hybrid (POM + Helper utilities)

 Something else

Skip

Q: What is your preferred project structure approach?
A: Hybrid (POM + Helper utilities)

Perfect! Now let me ask the second question.
Please select your preferred reporting format.



Which reporting format would you prefer?



1 Allure



2 HTML (built-in)

3 Monocart

4 Multiple reporters (HTML + JSON)

Something else

Skip

Q: Which reporting format would you prefer?
A: Multiple reporters (HTML + JSON)

Great choice! Moving to the third question.
Please select your preferred environment configuration approach.



How do you want to handle environment configuration?



1 dotenv only



2 Multiple environments (dev, staging, prod)

3 Environment-specific config files

4 All of the above

Something else

Skip

Q: How do you want to handle environment configuration?
A: Multiple environments (dev, staging, prod)

Excellent! Now for the fourth question.
Please select your preferred logging mechanism.



What logging mechanism would you prefer?



1 Winston



2 Pino

3 Built-in console with formatting

4 Custom logger utility

Something else

Skip

Q: What logging mechanism would you prefer?
A: Built-in console with formatting

Perfect! Now for the final question.
Please select the type of application scenarios for your test cases.



What type of application scenarios should the test cases cover?



1 Basic (login, search, form)



2 E-commerce (cart, checkout, products)

3 Dashboard/Admin panel

4 Custom - I'll specify later

Something else

Skip

Perfect! 🎉 I've successfully created a comprehensive Playwright TypeScript framework for you! Here's what has been delivered:

✅ **Framework Summary**

Architecture & Structure

- ✅ **Hybrid Pattern:** Page Object Model (POM) + Helper Utilities
- ✅ **TypeScript:** Full type safety throughout
- ✅ **Multi-Environment:** Dev, Staging, Production configurations
- ✅ **Multiple Reporters:** HTML, JSON, JUnit, and List

Test Coverage (70 Total Test Cases)

UI Tests (40 Test Cases)

1. **Authentication** (TC001-TC010): Login, logout, security tests
2. **Products** (TC011-TC020): Display, search, filter, sort, add to cart
3. **Shopping Cart** (TC021-TC030): Add, remove, update, promo codes
4. **Checkout** (TC031-TC040): Complete purchase flow, validation, end-to-end

API Tests (30 Test Cases)



1. **Auth API** (API_TC001-API_TC010): Login, register, token validation

API Tests (30 Test Cases)

1. **Auth API** (API_TC001-API_TC010): Login, register, token validation
2. **Products API** (API_TC011-API_TC020): CRUD operations, search, filter
3. **Cart API** (API_TC021-API_TC030): Cart management, validation

Key Features

- 🎯 Custom Logger with color-coded output
- 🛠 Data Generator for realistic test data
- ⚡ Parallel test execution
- 📸 Screenshots & videos on failure
- 🌐 Cross-browser testing (Chrome, Firefox, Safari)
- 📱 Mobile testing support
- 🔄 CI/CD ready (GitHub Actions workflow included)

Framework Components

- **13 Page Objects** including base page
- **4 Helper Utilities** (API, Wait, Assertion)
- **2 Core Utilities** (Logger, Data Generator)
- **Test Fixtures** with reusable test data



- **Type Definitions** for type safety

Documentation Included

-  Comprehensive README.md
-  Quick Start Guide
-  Test Execution Guide
-  Contributing Guidelines
-  GitHub Actions CI/CD workflow

Download Your Framework

The complete framework is ready for download as **playwright-ecommerce-framework.zip** (42KB)

Next Steps:

1. Extract the ZIP file
2. Run `npm install`
3. Run `npm run install:browsers`
4. Configure your environment in `.env`
5. Start testing with `npm test`

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All test cases cover both **positive and negative scenarios**, including edge cases, security tests (SQL injection), and complete end-to-end workflows!



Playwright ecommerce framework
ZIP

 Open in Windows Explorer

